Paper Main Framming

While prior work focused on architectural innovation to overcome performance plateau, we demonstrate these plateaus, we demonstrate these plateaus stem from the underlaying data overlap, not model limitations.

Introduction

Deep Learning achieves around 90% to 95% accuracy on COVID-19 CT classification, yet physics-based approaches plateau at 70% , this 20% gap raises a fundamental question: Is this a limitation of the model or the limitation of data

We adopt a Data centric lens, analysing whether the ceiling reflects insufficient feature engineering or fundamental class overlap, Using ARSIVAE with improved training on the MosMeddataset ((5,324 slices) we discover:

1) Information overlap: Foundation models rediscover physics features with 90.6$ redundancy, explaining the 70% ceiling
2) Non-Linear gaps: Remaining 20% is mainly the spatial interactions absent in the handcrafter features

Related Work

- COVID-19 detection: CNN achieve 95% to 98% accuracy with post hoc explainability via GRAD-CAM and SHAP, however these models highlight where the models look and not what physics they use
- Physics Informed Learning: HU based features achieve around 70% accuracy, Attribute VAE aligns latent dimensions with attributes via supervised loss , originally for facial features , We adapt this with improved annealing for medical physical features
- Data Centric AI: No prior work quantifies the information celling imposed by the feature space in medical imaging
- 1 SOTA table comparison (PINN vs Ours vs CNN vs Post HOC based)

Method (mainly consist of - model, sanity checks and some dataset attributes

- MosMeddataset:
    - 5000 CT are condensed from the dataset ((CT_0: Normal, CT_1: Mild COVID, CT_2: Moderate COVID)). We validate quality 9 checks table(here) summarize them here
    - Class balance using the chi squares test to be named
    - Derive the 14 features via papers and the statistical significance here [14 papers of each feature]
    - All 14 physics features show complete statistical significance ($p < 0.0001$)
    - Dataset validation summary table insert here
- ARSIVAE (Attribute regularised soft interpretable Vae WITH Improved training)
    - Use the standard architecture [cite the paper here] : use a 85-dim latent space, encoder and decoder CNN's and attribute predictor, due to the need for physics alignment and also decent reconstruction a custom schedular was used for avoiding Posterior collapse

- Loss function used (MSE based reconstructions, Clamped KL and MSE attribute)
- Three phase annealing in detail a small table to talk about each phase
- Low Beta early prevents posterior collapse, letting latent space organize the physics, gradual increase in lambda increases the alignment after the reconstruction stabilizes
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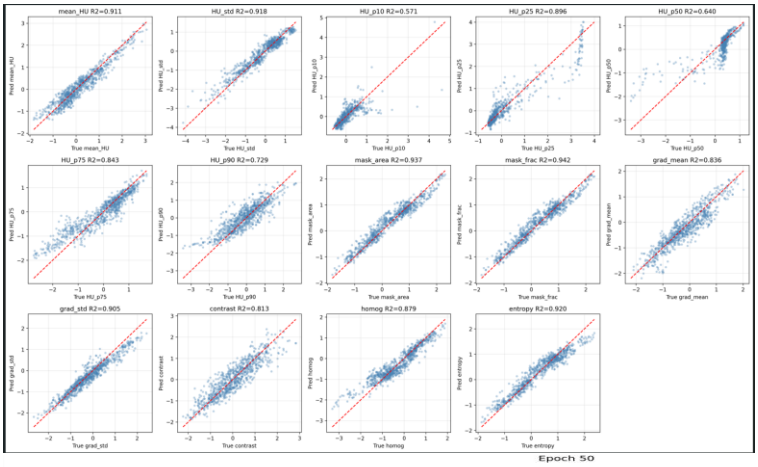
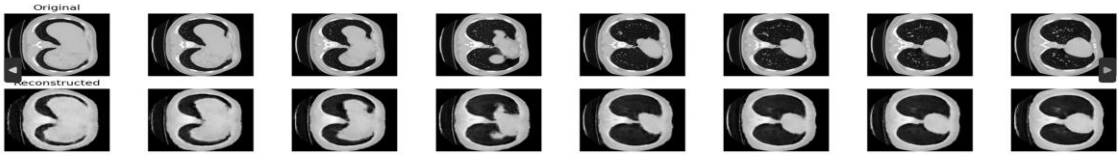
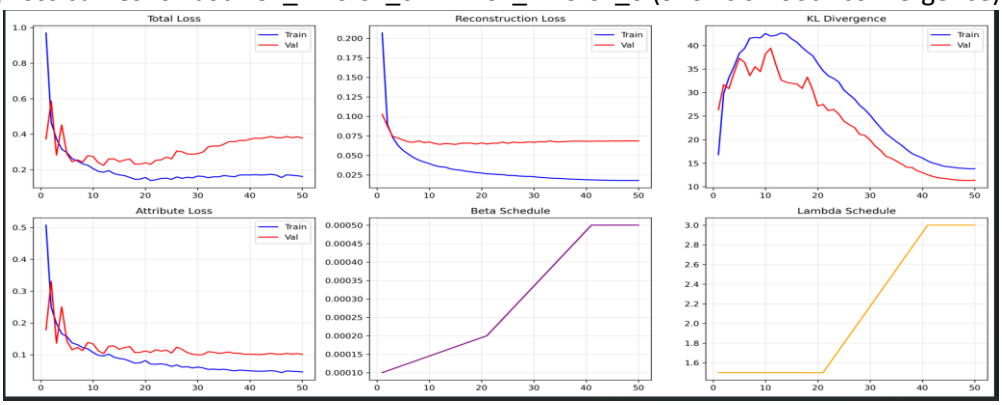
- Evaluation
  - Physics are aligned via R^2 per features. Classification via logistic regression on the 85D latent space, the model ran on 3 random seeds
  - Talk about reconstruction
  - Talk about Latent space and physics alignment figures of each with the R^2 scores from the figure obtained
  - 

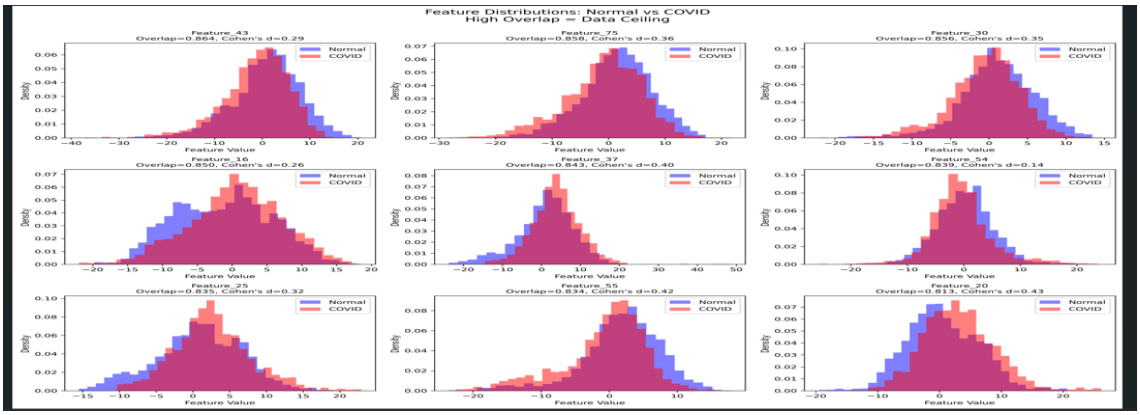
  - 

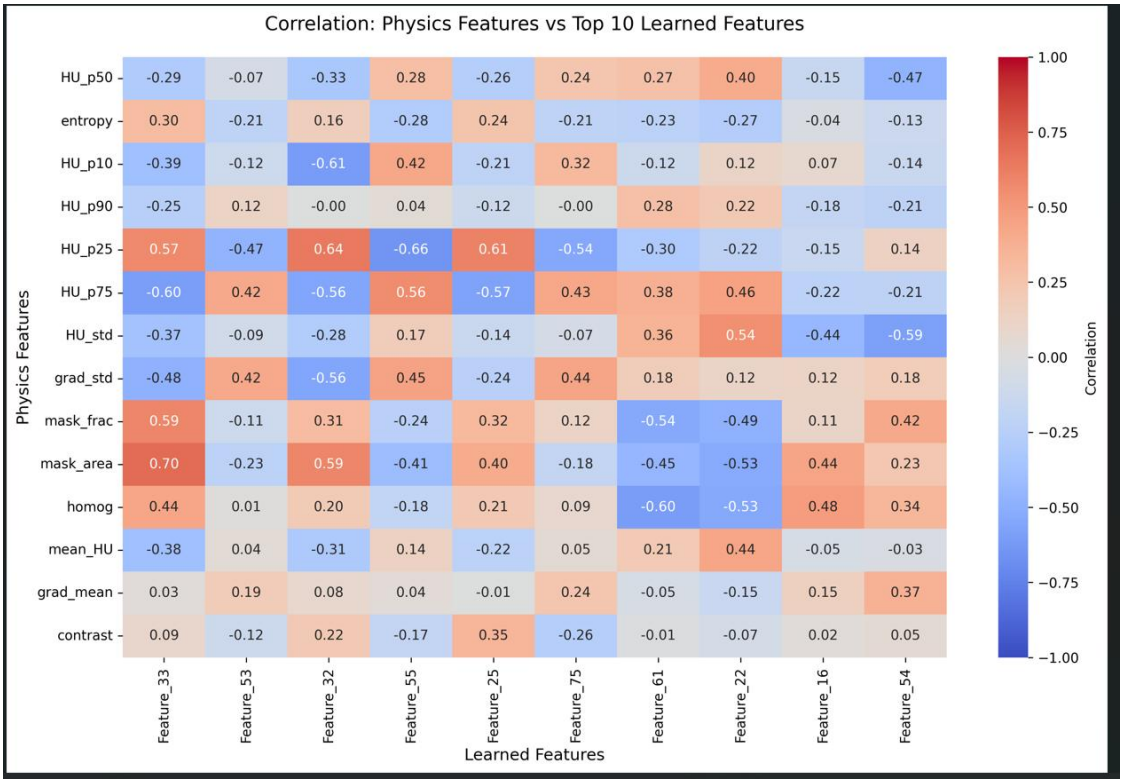
  - Overlap analysis done on the 71 learned features (pretrained VAE) and 14 derived features
  - A Cohen's D test done as well on the dataset
  - Results are available for both CT_0 VS CT_1 AND CT_2 VS CT_0

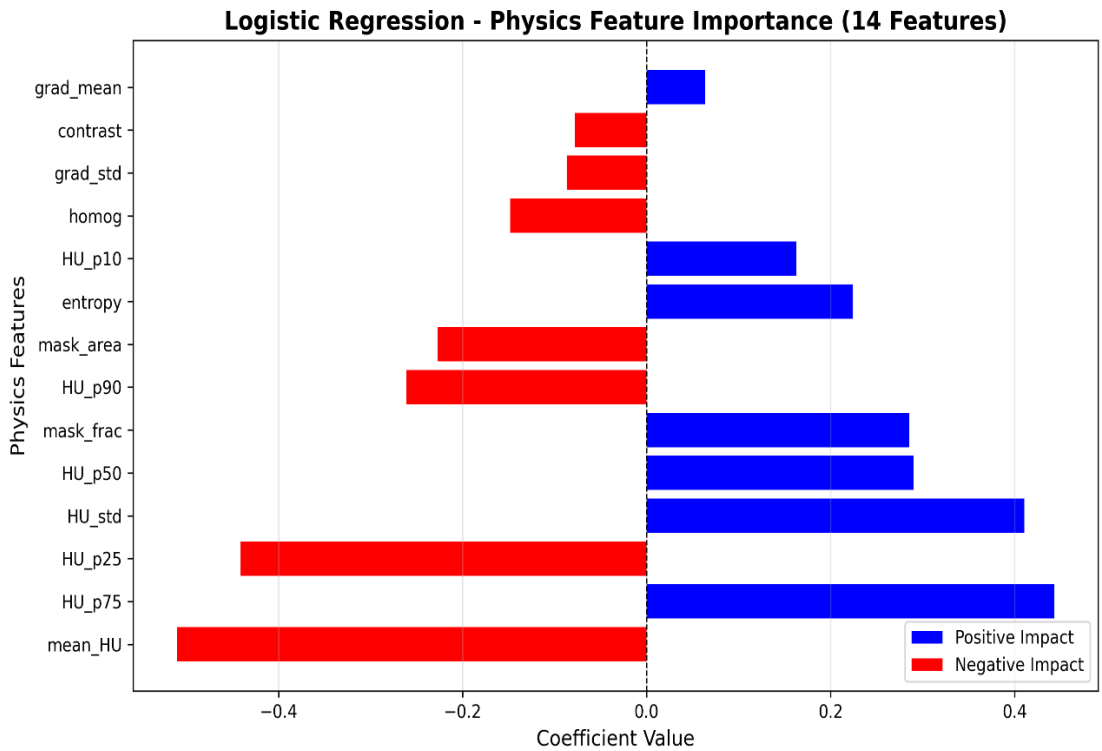
Results:

- Training Loss curves for both CT_1 VS CT_0 AND CT_2 VS CT_0 (Shows smooth convergence)
  - 

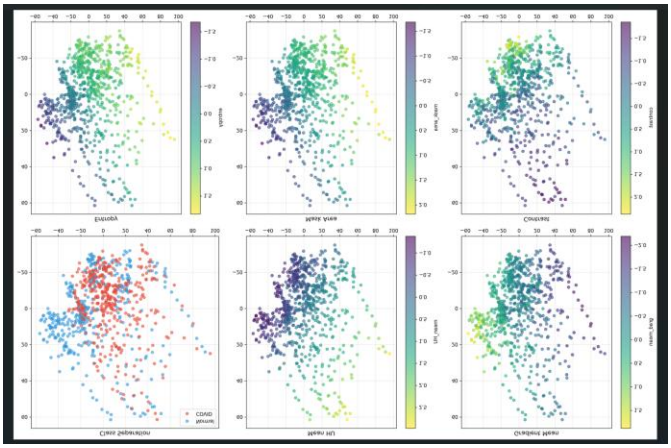
- Physics alignment referrers to the table with 3 seeded each for CT_1 VS CT_0 and then the CT_2 vs CT_0

- Table 2 for the best R^2 get the R^2 value for each category for CT_1 VS CT_0 and CT_2 vs CT_0
- Classification results amongst the models 1 each from the CT1 AND CT2
    - R^2 of CT higher but has lower separability
        - Reason: R^2 measures the feature reconstruction in the latent space, not class separability. CT_1 physics features overlap with normal (Cohens'd = x value) while CT_2 features are more well separated than CT_1 [refer to a figure] [cohens d value]
- Quantifying the Information ceiling
    - From correlation analysis from 71 learned features and the 14 radiological features reveals a 80.6% mean overlap
    - 

- 

- 

- Feature Interaction analysis
  - We use logistic regression to get the feature interaction via persons correlation
  - 80% of the learned features are redundant with physics combinations, remaining 20% captures non-linear interactions that handcrafter features miss
- Use the PCA plot here to show the separability matching the classification accuracy, Physics features vary smoothly across manifold, confirming the physics grounded organizations

- 
Chlorophyll Fluorescence
Chlorophyll
Leaf Area
Leaf Area Index
Leaf Mass
Leaf Mass Index
Leaf Nitrogen
Leaf Nitrogen Index
Leaf Water
Leaf Water Index
Leaf Water Content
Leaf Water Content Index
Leaf Water Potential
Leaf Water Potential Index
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